

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B.Tech.

Branch : Electronics Engineering

Semester : III

Subject Code & Name: BTEXC303_Y18 Electronic Devices & Circuits

Max Marks: 60

Date: 23-07-2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 The region in which JFET operates as an amplifier is: a) Cut-off region b) Ohmic region c) Saturation region d) Breakdown region	CO 3	1
2 The input impedance of a JFET is: a) Low b) Medium c) High d) Very low	CO 3	1
3 Which of the following MOSFET types is normally off? a) D-MOSFET b) E-MOSFET c) JFET d) UJT	CO 3	1
4 A MOSFET can be used as a: a) Diode b) Amplifier c) Switch d) All of the above	CO 4	1
5 In CMOS inverter, power dissipation mainly occurs: a) During switching b) During idle c) When output is high d) When input is low	CO 4	1
6 An amplifier with negative feedback exhibits: a) Improved bandwidth b) Increased gain c) Increased noise d) Poor stability	CO 4	1
7 Cascade amplifier is used to: a) Increase input resistance b) Increase gain c) Increase output resistance d) Decrease gain	CO 4	1
8 DC amplifiers are used to amplify: a) Low-frequency AC b) DC signals c) High-frequency signals d) Noise	CO 2	1
9 Which of the following is an RC oscillator? a) Hartley b) Colpitts c) Wein Bridge d) Crystal	CO 4	1
10 The oscillator used for frequency stability is: a) Colpitts b) Crystal c) Hartley d) Phase shift	CO 4	1
11 Which oscillator uses FET and phase shift network? a) Hartley b) Wien c) RC Phase Shift d) Relaxation	CO 4	1
12 555 Timer has how many comparators? a) 1 b) 2 c) 3 d) 4	CO 2	1

Q. 2	Solve the following.		12
A)	Derive the expression for drain current I_D in the saturation region of an n-channel JFET.	CO 1	6
B)	Compare CS, CD, and CG configurations of JFET.	CO 2	6
Q.3	Solve the following.		12
A)	Explain the working of CMOS inverter and describe its VTC (Voltage Transfer Characteristic).	CO 2	6
B)	Draw and explain the CMOS inverter circuit and its operation.	CO 2	6
Q. 4	Solve Any Two of the following.		12
A)	Explain the characteristics and advantages of negative feedback in amplifiers.	CO 3	6
B)	Describe the role and working of a DC amplifier.	CO 3	6
C)	What is meant by low-noise amplifier? Where is it used and why?	CO 4	6
Q.5	Solve Any Two of the following.		12
A)	Explain Barkhausen criterion and how it ensures sustained oscillations.	CO 3	6
B)	Derive an equation for frequency of Colpitts oscillator.	CO 4	6
C)	Explain the operation of UJT relaxation oscillator with circuit and waveform.	CO 2	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the internal block diagram of IC555 timer.	CO 2	6
B)	Describe the working of bistable multivibrator using FETs.	CO 2	6
C)	Describe how a monostable multivibrator can be used as a timer in practical circuits.	CO 2	6

***** End *****